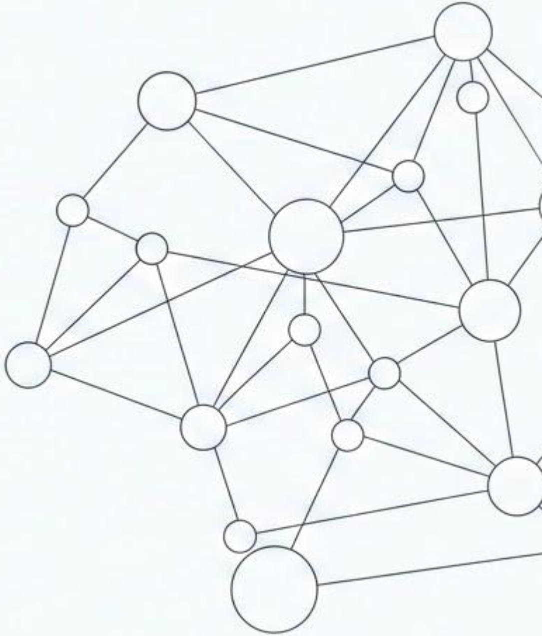
Graph-Augmented RAG (G-RAG)

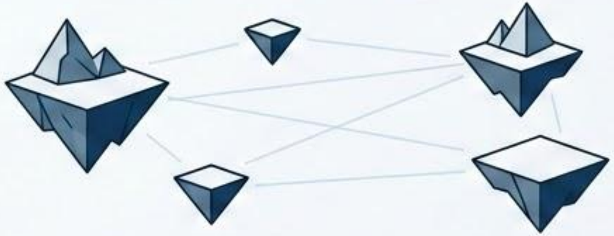
When Relationships Matter More Than Similarity



Limitations of Vector-Only RAG

- Good for similarity, poor for relationships
- Weak at multi-hop reasoning
- Loses structured connections
- Hard to do complex queries

Disconnected Islands vs Connected Graph



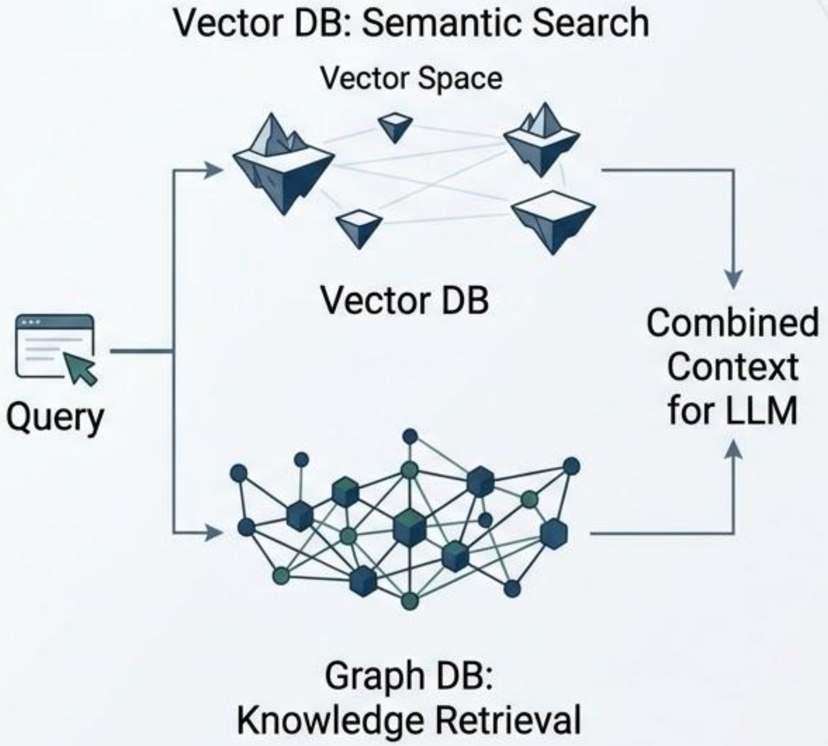
Vectors: Similarity Only

Graph: Complex Relationships

What is Graph-Augmented RAG?

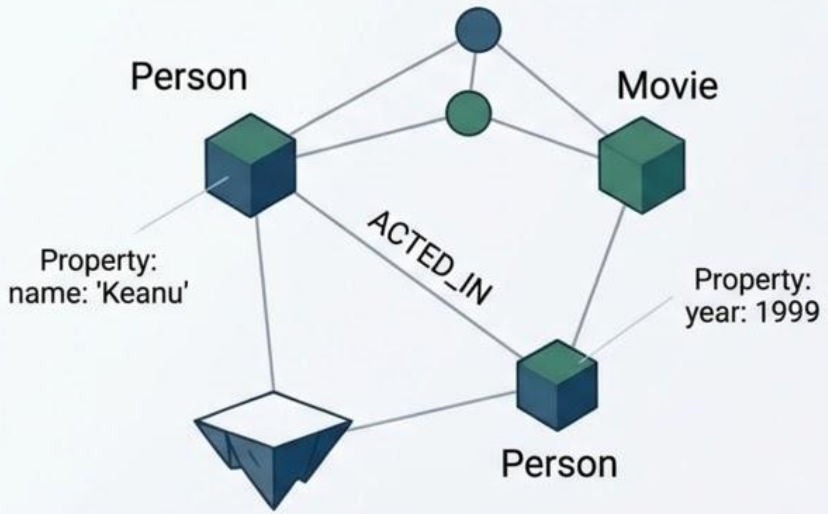
Definition: Combining Knowledge Graphs with Vector Search

Benefits: Structured reasoning + semantic search



Neo4j Fundamentals

- Nodes (Entities)
- Relationships (Connections)
- Properties (Attributes)
- Labels (Types)



Cypher Query Language

Key Clauses:

- CREATE: Inserting data
- MATCH: Finding patterns
- RETURN: Specifying results

Simple Queries Shown

```
MATCH (n:Person)
RETURN n

CREATE (p:Person {name: 'Alice'})

...
MATCH (n)
RETURN n
```

Entity & Relationship Extraction

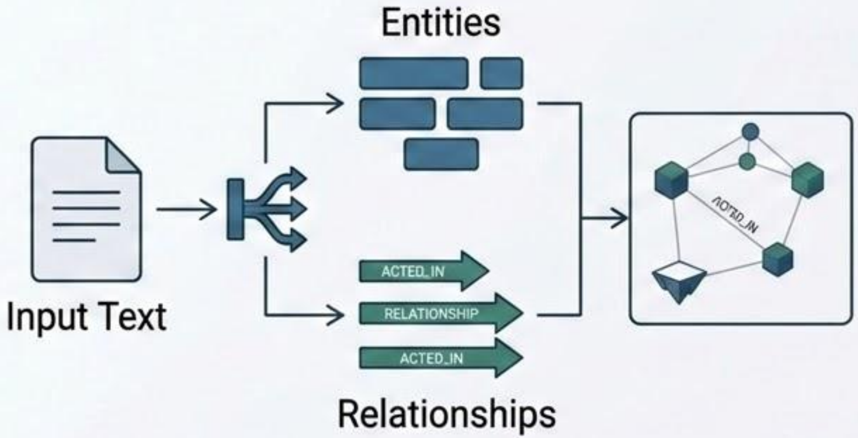
Processing Workflow

Text → Entities →

Relationships → Graph

Extraction Tools

- LLMs
- spaCy spaCy
- LangChain

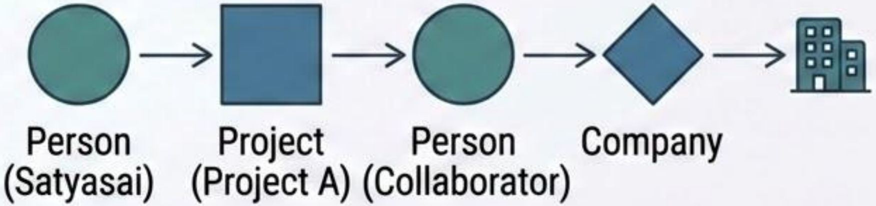


When Graph RAG Beats Vector Search

USE CASE	COMPARISON	BEST FOR
Multi-hop reasoning (A → B → C)	Excelled with explicit entity relationships in Knowledge Graphs.	✓ Graph RAG
Structured domain knowledge	Preserves complex logical schemas and hierarchies effectively.	✓ Graph RAG
Recommendation & dependency analysis	Identifies hidden patterns and deep-link influerces via graph traversal.	✓ Graph RAG
Compliance & audit trails	Maintains clear, explainable data lineage and provenances.	✓ Graph RAG

Multi-Hop Reasoning in Graphs

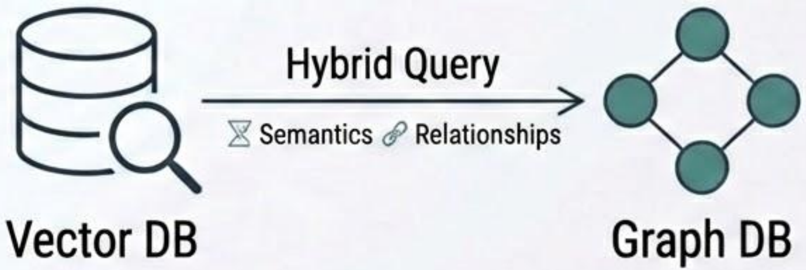
- Multi-Hop Reasoning
  - Connecting multiple entities to answer complex questions
  - Understanding indirect relationships
  - Example Query: "Who worked on projects with Satyasai?"
  - Example Query: "What regulations affect our AI products?"



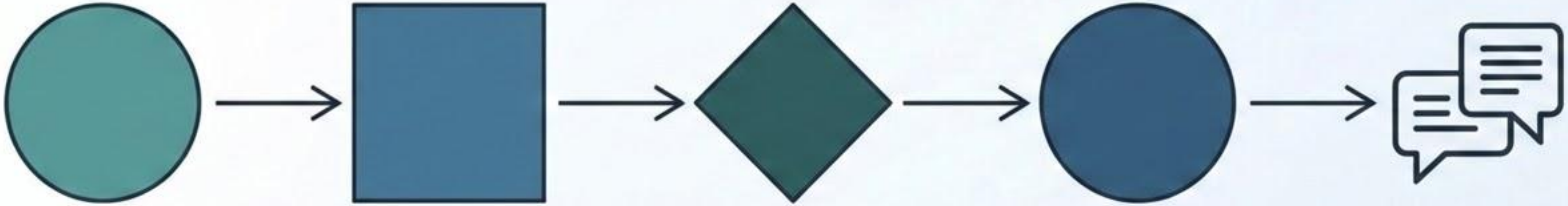
Hybrid Vector + Graph Approach

🏢 **Architecture: Vector for Semantics + Graph for Relationships**

- Benefits of Both Worlds
  - Enhanced Accuracy & Coverage
  - Structured, Explainable Results
  - Comprehensive Insights



Graph RAG Pipeline



1. Entity Extraction
2. Relationship Building
3. Graph Storage
4. Hybrid Retrieval
5. Answer Generation

Best Practices for Graph RAG

- Start small, scale gradually
- Keep graph schema clean
- Use hybrid approach
- Maintain graph freshness
- Monitor query performance

Key Takeaways

- Graphs excel at relationships and multi-hop reasoning
- Use Graph RAG when structure matters
- Hybrid (Vector + Graph) is often the best approach
- Neo4j + Cypher is powerful and practical

Use the right tool for the right job